EXHAUST PIPE (Continued)

INSTALLATION - 4.0L

(1) Position exhaust pipe and catalytic converter into vehicle.

(2) Insert exhaust pipe hanger into transmission support (Fig. 6).

(3) Install exhaust pipe onto exhaust manifold **DO NOT** tighten bolts at this time.

(4) Position muffler flange onto catalytic converter flange and install retaining bolts and nuts (Fig. 6). **DO NOT** tighten nuts at this time.

(5) Make sure the exhaust system is aligned and has the proper clearance. The minimum clearance is 25mm (1 inch).

(6) Tighten muffler to catalytic converter flange retaining nuts to 28.5 N·m (21 ft. lbs.).

(7) Tighten exhaust pipe to exhaust manifold mounting bolts to 31 N·m (23 ft. lbs.).

(8) Install the oxygen sensors in the exhaust pipe and catalytic converter.

(9) Lower vehicle.

(10) Start engine and inspect for leaks. Repair exhaust leaks as necessary.

(11) Check the exhaust system for contact with the body panels. Make the necessary adjustments, if needed.

HEAT SHIELDS

DESCRIPTION

Heat shields (Fig. 7) are made of stamped/formed steel, or metal foil.

Exhaust heat shields are needed to protect both the vehicle and the environment from the high temperatures developed by the catalytic converter. The catalytic converter releases additional heat into the exhaust system. Under severe operating conditions, the temperature increases in the area of the converter. Such conditions can exist when the engine misfires or otherwise does not operate at peak efficiency.

REMOVAL

(1) Raise and support the vehicle.

(2) Remove the screws and/or nuts holding the heat shields to the frame and/or floor pan (Fig. 8) and (Fig. 9).

(3) When removing muffler heat shield, the muffler front support bracket must be removed first.

(4) Slide the shields out around the exhaust system.

INSTALLATION

(1) Position the heat shields to the floor pan or the frame and install the screws and/or nuts (Fig. 8) and (Fig. 9).

(2) Tighten the nuts and/or screws to 45 N·m (33 ft. lbs.).

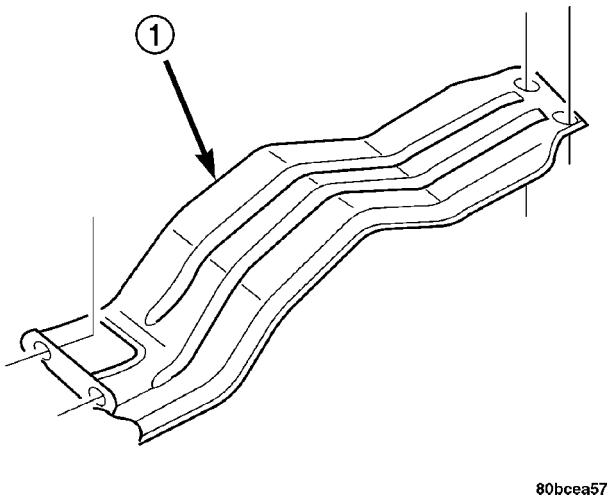

Fig. 7 Exhaust Heat Shield - Typical

1 - HEAT SHIELD

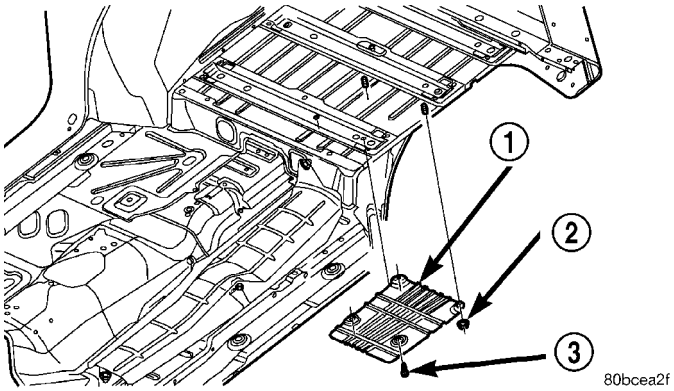

Fig. 8 Muffler Heat Shield - Rear

1 - MUFFLER HEAT SHIELD - REAR
2 - NUTS
3 - SELF TAPPING SCREWS

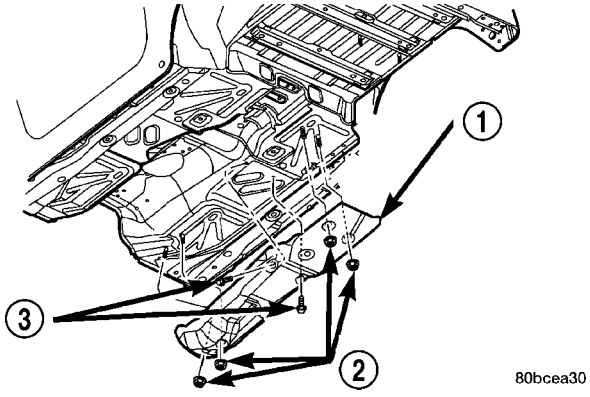

Fig. 9 Muffler Heat Shield - Middle

1 - MUFFLER HEAT SHIELD - MIDDLE
2 - NUTS
3 - SELF TAPPING SCREWS

(3) Lower the vehicle.